

Time Series Solutions to Mock Exam

1. (a) The time series $\{X_t\}$ is said to be second-order/weak or covariance stationary if for all $n \geq 1$ for any $t_1, \dots, t_n \in \mathbb{Z}$ and for all τ such that $t_1 + \tau, \dots, t_n + \tau \in \mathbb{Z}$ all the joint moments of order 1 and 2 of $X_{t_1}, \dots, X_{t_n}$ exist, are all finite and equal to the corresponding joint moments of $X_{t_1+\tau}, \dots, X_{t_n+\tau}$. Equivalently, $\forall t, s \in \mathbb{Z}, \forall \tau$ such that $t + \tau, s + \tau \in \mathbb{Z}$

1. $\mathbb{E}[X_t] = \mu,$
2. $\text{Var}(X_t) = \sigma^2 < \infty,$
3. $\mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[X_s X_{s+\tau}].$

The time series $\{X_t\}$ is said to be completely/strong or strictly stationary if for all $n \geq 1$ for any $t_1, \dots, t_n \in \mathbb{Z}$ and for all τ such that $t_1 + \tau, \dots, t_n + \tau \in \mathbb{Z}$ the joint distribution of $X_{t_1}, \dots, X_{t_n}$ is the same as $X_{t_1+\tau}, \dots, X_{t_n+\tau}$.

- (b) An ARMA(p,q) process is specified by

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \epsilon_t - \theta_1 \epsilon_{t-1} - \dots - \theta_q \epsilon_{t-q},$$

where ϕ_i and θ_j are constants for $i = 1, \dots, p$, and $j = 1, \dots, q$, and $\{\epsilon_t\}$ is a white noise process.

- (c) We have

$$\begin{aligned} \text{cov}(X_t, X_{t+\tau}) &= \mathbb{E}[X_t X_{t+\tau}] - \mathbb{E}[X_t] \mathbb{E}[X_{t+\tau}] \\ &= \mathbb{E}[(\epsilon_t - 0.2\epsilon_{t-1} + 0.1\epsilon_{t-2})(\epsilon_{t+\tau} - 0.2\epsilon_{t+\tau-1} + 0.1\epsilon_{t+\tau-2})] \\ &= \mathbb{E}[\epsilon_t \epsilon_{t+\tau} - 0.2\epsilon_t \epsilon_{t+\tau-1} + 0.1\epsilon_t \epsilon_{t+\tau-2} \\ &\quad - 0.2\epsilon_{t-1} \epsilon_{t+\tau} + 0.04\epsilon_{t-1} \epsilon_{t+\tau-1} - 0.02\epsilon_{t-1} \epsilon_{t+\tau-2} \\ &\quad + 0.1\epsilon_{t-2} \epsilon_{t+\tau} - 0.02\epsilon_{t-2} \epsilon_{t+\tau-1} + 0.01\epsilon_{t-2} \epsilon_{t+\tau-2}] \\ &= \mathbb{E}[\epsilon_t \epsilon_{t+\tau}] - 0.2\mathbb{E}[\epsilon_t \epsilon_{t+\tau-1}] + 0.1\mathbb{E}[\epsilon_t \epsilon_{t+\tau-2}] \\ &\quad - 0.2\mathbb{E}[\epsilon_{t-1} \epsilon_{t+\tau}] + 0.04\mathbb{E}[\epsilon_{t-1} \epsilon_{t+\tau-1}] - 0.02\mathbb{E}[\epsilon_{t-1} \epsilon_{t+\tau-2}] \\ &\quad + 0.1\mathbb{E}[\epsilon_{t-2} \epsilon_{t+\tau}] - 0.02\mathbb{E}[\epsilon_{t-2} \epsilon_{t+\tau-1}] + 0.01\mathbb{E}[\epsilon_{t-2} \epsilon_{t+\tau-2}] \\ &= \begin{cases} 0.1\sigma^2 & \text{if } \tau = -2, \\ -0.22\sigma^2 & \text{if } \tau = -1, \\ 1.05\sigma^2 & \text{if } \tau = 0, \\ -0.22\sigma^2 & \text{if } \tau = 1, \\ 0.1\sigma^2 & \text{if } \tau = 2, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

2. (a) Write

$$a = 1 + 0.04^2, \quad \varphi = \frac{0.5}{a}.$$

The model can be written as

$$(a - 0.5B)X_t = (1 + 0.25B)\epsilon_t,$$

or equivalently

$$(1 - \varphi B)X_t = \frac{1 + 0.25B}{a}\epsilon_t.$$

Since $|\varphi| < 1$, we have

$$\frac{1}{1 - \varphi B} = \sum_{j=0}^{\infty} \varphi^j B^j.$$

Therefore

$$\begin{aligned} X_t &= \frac{1}{a} \left(\sum_{j=0}^{\infty} \varphi^j B^j \right) (1 + 0.25B)\epsilon_t \\ &= \frac{1}{a} \sum_{j=0}^{\infty} \varphi^j \epsilon_{t-j} + \frac{0.25}{a} \sum_{j=0}^{\infty} \varphi^j \epsilon_{t-j-1}. \end{aligned}$$

Hence

$$X_t = \sum_{j=0}^{\infty} \lambda_j \epsilon_{t-j},$$

where

$$\lambda_0 = \frac{1}{a}, \quad \lambda_j = \frac{1}{a} (\varphi^j + 0.25\varphi^{j-1}), \quad j \geq 1.$$

Equivalently, for $j \geq 1$,

$$\lambda_j = \frac{\varphi + 0.25}{a} \varphi^{j-1}.$$

(b) Since $X_t = \sum_{j=0}^{\infty} \lambda_j \epsilon_{t-j}$, the autocovariance sequence is

$$\gamma_\tau = \text{Cov}(X_{t+\tau}, X_t) = \sigma_\epsilon^2 \sum_{j=0}^{\infty} \lambda_j \lambda_{j+|\tau|}.$$

Using the coefficients from part (a), define

$$b = \frac{\varphi + 0.25}{a}.$$

Then $\lambda_0 = 1/a$ and $\lambda_j = b\varphi^{j-1}$ for $j \geq 1$. Therefore

$$\gamma_0 = \sigma_\epsilon^2 \left\{ \frac{1}{a^2} + \frac{b^2}{1 - \varphi^2} \right\},$$

and for $\tau \geq 1$,

$$\gamma_\tau = \sigma_\epsilon^2 \left\{ \frac{b}{a} \varphi^{\tau-1} + \frac{b^2 \varphi^\tau}{1 - \varphi^2} \right\}.$$

Finally, by symmetry,

$$\gamma_{-\tau} = \gamma_\tau.$$

(c) One way to predict X_{n+1} is to first reconstruct the past innovations recursively. From

$$(a - 0.5B)X_t = (1 + 0.25B)\epsilon_t,$$

we obtain

$$\epsilon_t = aX_t - 0.5X_{t-1} - 0.25\epsilon_{t-1}.$$

Starting from initial values such as $X_0 = 0$ and $\epsilon_0 = 0$, this recursion gives estimates $\hat{\epsilon}_1, \dots, \hat{\epsilon}_n$. Using the truncated infinite-order representation from part (a),

$$X_{n+1} \approx \sum_{j=0}^5 \lambda_j \epsilon_{n+1-j}.$$

Since ϵ_{n+1} is unpredictable from the observed past and has mean zero,

$$\hat{X}_{n+1} = \mathbb{E}[(X_{n+1} \mid X_n, \dots, X_1)] \approx \sum_{j=1}^5 \lambda_j \hat{\epsilon}_{n+1-j}.$$

Equivalently, using the ARMA recursion directly gives the one-step forecast

$$\hat{X}_{n+1} = \frac{0.5X_n + 0.25\hat{\epsilon}_n}{a}.$$

3. (a) We first rewrite $\hat{S}_p(f)$ as

$$\hat{S}_p(f) = \frac{1}{N} \sum_{t_1=1}^N \sum_{t_2=1}^N X_{t_1} X_{t_2} \exp\{-2i\pi f(t_1 - t_2)\}$$

Applying the expectation operator $\mathbb{E}$, and noting that $\text{Cov}(X_{t_1}, X_{t_2}) = \mathbb{E}(X_{t_1} X_{t_2})$ (mean is zero) we obtain

$$\mathbb{E}(\hat{S}_p(f)) = \frac{1}{N} \sum_{t=1}^N \mathbb{E}(X_t X_t) = \text{Var}(X_t) = \sigma^2$$

(b) We rewrite $\hat{S}_p(f)$ again as

$$\hat{S}_p(f) = \frac{1}{N} \sum_{\tau=-N+1}^{N-1} \sum_{t=1}^{N-|\tau|} X_t X_{t+\tau} \exp\{-2i\pi f\tau\} = \sum_{\tau=-N+1}^{N-1} \hat{\gamma}_X^{(p)}(\tau) \exp\{-2i\pi f\tau\}$$

Integrating, and using linearity of the integral operator in the first passage, we compute

$$\begin{aligned} \int_{-1/2}^{1/2} \hat{S}_p(f) df &= \sum_{\tau=-N+1}^{N-1} \int_{-1/2}^{1/2} \hat{\gamma}_X^{(p)}(\tau) \exp\{-2i\pi f\tau\} df \\ &= \sum_{\tau=-N+1}^{N-1} \hat{\gamma}_X^{(p)}(\tau) \int_{-1/2}^{1/2} \exp\{-2i\pi f\tau\} df \\ &= \sum_{\tau=-N+1}^{N-1} \hat{\gamma}_X^{(p)}(\tau) \int_{-1/2}^{1/2} \cos\{-2\pi f\tau\} df \\ &= \hat{\gamma}_X^{(p)}(0) + \sum_{\tau=-N+1 \setminus \{0\}}^{N-1} \hat{\gamma}_X^{(p)}(\tau) \frac{\sin\{-\pi\tau\} - \sin\{\pi\tau\}}{-2\pi\tau} \\ &= \hat{\gamma}_X^{(p)}(0) \end{aligned}$$

In the third passage we have used Euler's formula, and the fact that sine is an odd function. Taking expectation we obtain,

$$\begin{aligned} \mathbb{E} \left[\int_{-1/2}^{1/2} \hat{S}_p(f) df \right] &= \mathbb{E} [\hat{\gamma}_X^{(p)}(0)] \\ &= \mathbb{E} \left[\frac{1}{N} \sum_{t=1}^N X_t X_t \right] \\ &= \text{Var}(X_t) \\ &= \sigma^2 \end{aligned}$$

(c) We have that $X_t = \epsilon_t + \theta\epsilon_{t-1}$ so therefore

$$\begin{aligned} \text{Cov}(X_{t+\tau}, X_t) &= \text{Cov}(\epsilon_{t+\tau} + \theta\epsilon_{t+\tau-1}, \epsilon_t + \theta\epsilon_{t-1}) \\ &= \text{Cov}(\epsilon_{t+\tau}, \epsilon_t) + \theta \text{Cov}(\epsilon_{t+\tau}, \epsilon_{t-1}) + \theta \text{Cov}(\epsilon_{t+\tau-1}, \epsilon_t) + \theta^2 \text{Cov}(\epsilon_{t+\tau-1}, \epsilon_{t-1}) \\ &= \sigma^2 \delta_{\tau,0} + \theta\sigma^2 \delta_{\tau+1,0} + \theta\sigma^2 \delta_{\tau-1,0} + \theta^2 \sigma^2 \delta_{\tau,0} \\ &= \begin{cases} \sigma^2(1 + \theta^2) & \text{if } \tau = 0, \\ \theta\sigma^2 & \text{if } |\tau| = 1, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

This is invariant to t , and finite, and the mean is zero, hence the process is second-order stationary. Now the autocovariance of $\{X_t\}$ is therefore given by

$$\gamma_\tau = \begin{cases} \sigma^2(1 + \theta^2) & \text{if } \tau = 0, \\ \theta\sigma^2 & \text{if } |\tau| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The spectral density function is then by definition given by

$$\begin{aligned} \mathcal{S}(f) &= \sum_{\tau=-\infty}^{\infty} \gamma_\tau e^{-2\pi i f \tau} \\ &= \theta\sigma^2 e^{2\pi i f} + (1 + \theta^2)\sigma^2 + \theta\sigma^2 e^{-2\pi i f} \\ &= \sigma^2(1 + \theta^2 + 2\theta \cos(2\pi f)). \end{aligned}$$

4. (a) Recall that $X_t^{(1)} = cX_{t-d}^{(2)} + \varepsilon_t$. We need to check three conditions for second-order stationarity. Firstly, we need to check that the mean is constant over time. We have for all $t \in \mathbb{Z}$

$$\begin{aligned}\mathbb{E}[X_t^{(1)}] &= \mathbb{E}[cX_{t-d}^{(2)} + \varepsilon_t] \\ &= c\mathbb{E}[X_{t-d}^{(2)}] + \mathbb{E}[\varepsilon_t] \\ &= 0,\end{aligned}$$

by assumption.

Secondly, we need to check that the variance is finite and constant over time. We have for all $t \in \mathbb{Z}$

$$\begin{aligned}\text{Var}(X_t^{(1)}) &= \text{Var}(cX_{t-d}^{(2)} + \varepsilon_t) \\ &= c^2 \text{Var}(X_{t-d}^{(2)}) + \text{Var}(\varepsilon_t) \\ &< \infty,\end{aligned}$$

again because both $X_t^{(2)}$ and ε_t are assumed to be second-order stationary.

Finally, we need to check that the covariance is constant over time. We have for all $t, s, \tau \in \mathbb{Z}$

$$\begin{aligned}\text{Cov}(X_{t+\tau}^{(1)}, X_t^{(1)}) &= \text{Cov}(cX_{t+\tau-d}^{(2)} + \varepsilon_{t+\tau}, cX_{t-d}^{(2)} + \varepsilon_t) \\ &= c^2 \text{Cov}(X_{t+\tau-d}^{(2)}, X_{t-d}^{(2)}) + \text{Cov}(\varepsilon_{t+\tau}, \varepsilon_t) \\ &= c^2 \text{Cov}(X_{s+\tau-d}^{(2)}, X_{s-d}^{(2)}) + \text{Cov}(\varepsilon_{s+\tau}, \varepsilon_s) \\ &= \text{Cov}(X_{s+\tau}^{(1)}, X_s^{(1)}),\end{aligned}$$

by stationarity of $X_t^{(2)}$ and ε_t .

- (b) We already know that both $X_t^{(1)}$ and $X_t^{(2)}$ are second-order stationary, so we need only check the cross-covariance is invariant to shifts in time. We have for all $t, s, \tau \in \mathbb{Z}$

$$\begin{aligned}\text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)}) &= \text{Cov}(cX_{t-d+\tau}^{(2)} + \varepsilon_{t+\tau}, X_t^{(2)}) \\ &= c \text{Cov}(X_{t-d+\tau}^{(2)}, X_t^{(2)}) + \text{Cov}(\varepsilon_{t+\tau}, X_t^{(2)}) \\ &= c \text{Cov}(X_{s-d+\tau}^{(2)}, X_s^{(2)}) + 0 \\ &= \text{Cov}(X_{s+\tau}^{(1)}, X_s^{(2)}),\end{aligned}$$

where we used the fact that ε_t is uncorrelated with $X_t^{(2)}$, and the stationarity of $X_t^{(2)}$.

- (c) Note that from the previous working (setting $s = 0$) we have

$$\begin{aligned}\gamma_\tau^{(1,2)} &= c\gamma_{\tau-d}^{(2,2)} + 0 = c\gamma_{\tau-d}^{(2,2)} \\ \gamma_\tau^{(2,1)} &= \gamma_{-\tau}^{(1,2)} = c\gamma_{-\tau-d}^{(2,2)} \\ \gamma_\tau^{(1,1)} &= c^2\gamma_\tau^{(2,2)} + \sigma_\varepsilon^2\delta_{\tau,0}.\end{aligned}$$

Write $S_{2,2}(f)$ for the spectral density of $X_t^{(2)}$ at $f \in \mathbb{R}$.

$$S_{2,2}(f) = \sum_{\tau=-\infty}^{\infty} \gamma_\tau^{(2,2)} e^{-2\pi i f \tau}$$

Then we have

$$\begin{aligned}
S_{1,2}(f) &= \sum_{\tau=-\infty}^{\infty} \gamma_{\tau}^{(1,2)} e^{-2\pi i f \tau} \\
&= \sum_{\tau=-\infty}^{\infty} c \gamma_{\tau-d}^{(2,2)} e^{-2\pi i f \tau} \\
&= c \sum_{\tau'=-\infty}^{\infty} \gamma_{\tau'}^{(2,2)} e^{-2\pi i f (\tau'+d)} \quad (\text{setting } \tau' = \tau - d) \\
&= c e^{-2\pi i f d} \sum_{\tau'=-\infty}^{\infty} \gamma_{\tau'}^{(2,2)} e^{-2\pi i f \tau'} \\
&= c e^{-2\pi i f d} S_{2,2}(f).
\end{aligned}$$

By symmetry we also have

$$\begin{aligned}
S_{2,1}(f) &= \sum_{\tau=-\infty}^{\infty} \gamma_{-\tau}^{(1,2)} e^{-2\pi i f \tau} \\
&= \sum_{\tau=-\infty}^{\infty} \gamma_{\tau'}^{(1,2)} e^{2\pi i f \tau'} \quad (\text{setting } \tau' = -\tau) \\
&= S_{1,2}(f)^* \\
&= c e^{2\pi i f d} S_{2,2}(f).
\end{aligned}$$

Finally, we have

$$\begin{aligned}
S_{1,1}(f) &= \sum_{\tau=-\infty}^{\infty} \gamma_{\tau}^{(1,1)} e^{-2\pi i f \tau} \\
&= \sum_{\tau=-\infty}^{\infty} \left(c^2 \gamma_{\tau}^{(2,2)} + \sigma_{\epsilon}^2 \delta_{\tau,0} \right) e^{-2\pi i f \tau} \\
&= c^2 \sum_{\tau=-\infty}^{\infty} \gamma_{\tau}^{(2,2)} e^{-2\pi i f \tau} + \sigma_{\epsilon}^2 \\
&= c^2 S_{2,2}(f) + \sigma_{\epsilon}^2.
\end{aligned}$$